DELHI PUBLIC SCHOOL JANKIPURAM, LUCKNOW

SESSION 2025-2026 : WORKSHEET-2

Subject: Science (086) Biology

Chapter: 1 – LIFE PROCESSES

Date: _____ Name : _____ Class / Sec. X_____

Section A: Multiple Choice Questions (MCQs)

(Choose the correct answer from the given options.)

1. The mode of nutrition in green plants is:

a. Heterotrophic b.Parasitic c.Autotrophic d.Saprophytic

2. Which of the following organisms use holozoic nutrition?

a. Mushroom b. Amoeba c. Green plants d.Fungi

3. The enzyme responsible for digestion of proteins in the stomach is:

a. Lipase b. Amylase c. Pepsin d. Trypsin

4. The main function of bile in digestion is to:

a. Break down carbohydrates b. Emulsify fats c. Digest proteins d. Absorb nutrients

5. The opening and closing of stomata depend on:

a. Temperature b. Water content of guard cells c. Presence of oxygen d. Carbon dioxide concentration

6. The blood vessel that carries oxygenated blood to the body is:

a. Pulmonary artery b. Pulmonary vein c. Aorta d. Vena cava

7. Which part of the nephron is responsible for filtration of blood?

a. Glomerulus b. Ureter c. Collecting duct d. Loop of Henle

8. The waste product of aerobic respiration in humans is:

a. Oxygen b. Lactic acid c. Carbon dioxide d.Water

9. Xylem is responsible for transporting:

a. Food b. Water c. Oxygen d. Enzymes

10. Which of the following is a part of the human excretory system?

a. Heart b. Liver c. Kidney d. Pancreas

Section B: Assertion-Reason Questions

(Choose the correct option: A) Both A and R are true, and R is the correct explanation of A. B) Both A and R are true, but R is not the correct explanation of A. C) A is true, but R is false. D) A is false, but R is true.)

11. **Assertion (A):** Photosynthesis occurs in green leaves.
Reason (R): Chlorophyll absorbs light energy to convert carbon dioxide and water into glucose.

12. **Assertion (A):** The heart pumps oxygen-rich blood to the body.
Reason (R): The left ventricle of the heart sends oxygenated blood through the arteries.

13. **Assertion (A):** Respiration is a process that releases energy in the form of ATP.
Reason (R): ATP is synthesized during glycolysis and oxidative phosphorylation.

14. **Assertion (A):** Veins carry oxygenated blood to the heart.
Reason (R): Veins have thinner walls and valves to prevent the backflow of blood.

15. **Assertion (A):** Excretion removes metabolic wastes from the body.
Reason (R): The kidney filters out harmful substances like urea and ammonia.

Section C: Activity-Based MCQs

(These questions involve real-life experiments and observations.)

16. A student places a few raisins in water. After a few hours, they swell up. What biological process has occurred?

a. Diffusion b. Osmosis c. Active transport d. Endocytosis

17. A gardener notices that plants in his field are wilting despite watering them regularly. What could be the possible reason?

a. Water is not reaching the roots b. Photosynthesis is not occurring

c. Excess transpiration is causing water loss d. Lack of minerals in soil

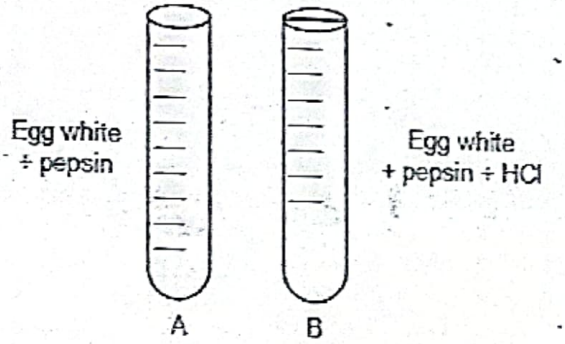
18. If red blood cells are placed in a highly concentrated salt solution, what will happen to them?

a. They will burst b. They will shrink due to water loss c. No change will occur d. They will absorb salt

Section D: Diagram-Based MCQs

(Identify the correct labels and structures based on the given diagrams.)

19. A student sets up an experiment to study the role of enzymes in digestion of food.



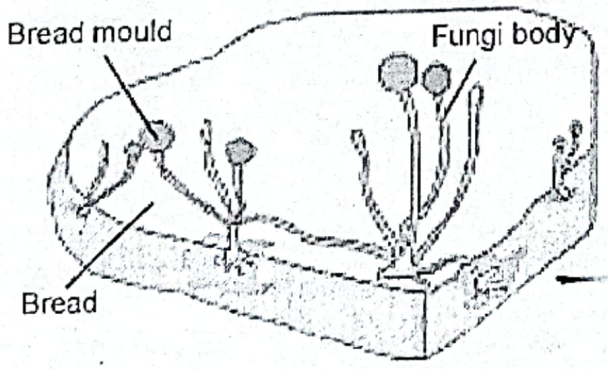
In which test tube, the digestion of protein will occur?
(a) Test tubes A as pepsin will breakdown protein into simple molecules.
(b) Test tube B as HCl will breakdown protein into simple molecules.
(c) Test tube A as pepsin will breakdown into simple molecules.
(d) Test tube B as HCl will activate pepsin for breakdown of protein into simple molecules.

20. Choose the correct path of urine in our body. [NCERT Exemplar]
(a) Kidney →ureter →urethra →urinary bladder (b) Kidney →urinary bladder →urethra →ureter
(c) Kidney →ureter →urinary bladder →urethra (d) Urinary bladder →kidney →ureter →urethra

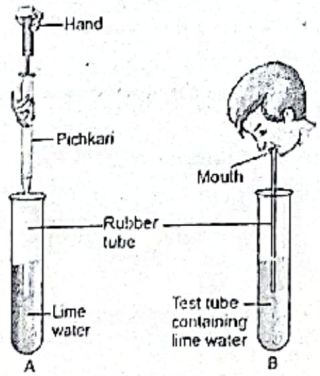
21. Which is the correct sequence of parts in human alimentary canal? [NCERT Exemplar]
(a) Mouth →stomach →small intestine →oesophagus →large intestine
(b) Mouth →oesophagus →stomach →large intestine →small intestine
(c) Mouth →stomach →oesophagus →small intestine →large intestine
(d) Mouth →oesophagus →stomach →small intestine →large intestine

22. If salivary amylase is lacking in the saliva, which of the following events in the mouth cavity will be affected? [NCERT Exemplar]
(a) Proteins breaking down into amino acids (b) Starch breaking down into sugars
(c) Fats breaking down into fatty acids and glycerol (d) Absorption of vitamins

23. The image shows the bread moulds on a bread. How these fungi obtain nutrition?
(a) By using nutrients from the bread to prepare their own food.
(b) By allowing other organisms to grow on the bread and then consuming them.
(c) By breaking down the nutrients of bread and then absorbing them.
(d) By eating the bread on which it is growing.

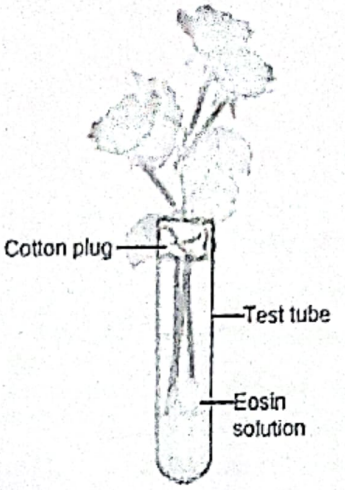


24. A student sets up an experiment to study human respiration using lime water, test tube and a straw. Lime water is colourless in the absence of CO_2 and is milky in its presence. The student fills a freshly prepared limewater in a test tube and blows air through straw into the limewater. It was observed that the solution turns cloudy as shown. What can be evaluated from this observation?



(a) Oxygen is exhaled during respiration. (b) Water vapours are produced during respiration.
(c) Carbon dioxide is exhaled during respiration. (d) Glucose is produced during respiration.

25. A student performs an experiment using a Balsam plant with intact stem, leaves, roots and flowers. The plant was kept in a test tube containing eosin solution (a pink colour dye). The test tube mouth was covered using cotton plug as shown.



The student kept the plant undisturbed in the lab. After 2-3 hours, a transverse section of stem was obtained using sharp scissors and studied under microscope. The studies reveal the presence of pink colour in the vessels of xylem. What does this observation explain?
(a) Most portion of the plant stem is occupied by xylem. (b) Water moves through xylem in the plant.
(c) Eosin solution gets stored in the xylem. (d) Xylem reacts with eosin and gives colour.

26. A few drops of iodine solution were added to rice water. The solution turned blue-black in colour. This indicates that rice water contains [NCERT Exemplar]
(a) complex proteins (b) simple proteins (c) fats (d) starch

27. Single circulation, i.e., blood flows through the heart only once during one cycle of passage through the body, is exhibited by [NCERT Exemplar]
(a) Labeo, Chameleon, Salamander (b) Hippocampus, Exocoetus, Anabas (c) Hyla, Rana, Draco (d) Whale, Dolphin, Turtle

28. Lack of oxygen in muscles often leads to cramps among cricketers. This results due to [NCERT Exemplar]
(a) conversion of pyruvate to ethanol (b) conversion of pyruvate to glucose
(c) non conversion of glucose to pyruvate (d) conversion of pyruvate to lactic acid

29. Rings of cartilage present in the throat ensure that
(a) air is filtered (b) air is at room temperature (c) air passage does not collapse (d) air is free of microbes

30. The inner lining of stomach is protected by one of the following from hydrochloric acid. Choose the correct one.
(a) Pepsin (b) Mucus (c) Salivary amylase (d) Bile

31. The opening and closing of the stomatal pore depends upon [NCERT Exemplar]
(a) oxygen (b) temperature (c) water in guard cells (d) concentration of CO_2 in stomata

Situation-Based MCQs :

32. A patient has been diagnosed with kidney failure and requires dialysis. What role does the dialysis machine play?
a. Absorbs nutrients from food b. Filters waste from the blood
c. Pumps oxygen into the bloodstream d. Transports blood from the lungs

33. A student places a potted plant in a room with only green light. Over time, the plant appears weak and unhealthy. What is the reason for this?

a. Green light reduces water absorption b. Green light is not efficiently used for photosynthesis

c. Green light increases the rate of respiration d. Green light prevents the plant from excreting oxygen

34. A doctor advises a patient to breathe deeply after experiencing dizziness. How will deep breathing help?

a. It reduces water loss through transpiration b. It increases oxygen supply to the body

c. It prevents digestion from slowing down d. It removes nitrogen waste from the blood

35. During a lab experiment, a student places an aquatic plant in sunlight. After a few minutes, bubbles start forming in the water around the plant. What process is responsible for bubble formation?

a. Respiration b. Photosynthesis c. Excretion d. Translocation

36. A person experiences muscle cramps during intense exercise. What causes this condition?

a. Excess oxygen in muscles b. Increased lactic acid due to anaerobic respiration

c. Reduced heart rate leading to low blood circulation d. Absence of protein digestion in muscles

37. A student conducts an experiment where a beaker containing iodine solution is covered with a semi-permeable membrane and placed in a starch solution. After some time, the iodine spreads throughout the starch solution. Which biological process is demonstrated here?

a. Osmosis b. diffusion c. active transport d. transpiration

38. A child accidentally ingests a large amount of sugary food. Soon after, their insulin levels rise. Why does this happen?

a. Insulin helps in protein digestion b. Insulin is necessary for blood clotting

c. Insulin regulates blood sugar levels d. Insulin transports oxygen in the blood

39. In an experiment, a student takes a fresh leaf and boils it in alcohol. After staining it with iodine solution, the leaf turns blue-black. What does this confirm?

a. The leaf contains proteins b. The leaf has performed respiration

c. The leaf contains starch from photosynthesis d. The leaf has absorbed iodine from alcohol

40. A person suffering from dehydration is advised to drink electrolyte-rich fluids. How will this help their cells?

a. It prevents excess oxygen intake b. It maintains the osmotic balance in cells

c. It absorbs nitrogen waste from cells d. It increases the digestion of food

41. A student places a raisin in pure water. After some time, it swells up. What biological process is responsible for this change?

a. Diffusion b. Active transport c. Osmosis d. Exocytosis

42. A researcher studies two plants: one placed in blue light and another in green light. The plant in blue light grows healthier. Why?

a. Blue light enhances water absorption b. Green light is not efficiently used in photosynthesis

c. Blue light prevents transpiration d. Green light increases oxygen intake

43. A student places a potato slice in concentrated salt solution and notices it shrinks. What process has occurred?

A. Diffusion b. Osmosis c. Transpiration d. Active transport

44. A doctor advises a dehydrated patient to drink electrolyte-rich fluids. How does this help their cells?

a. It reduces oxygen levels in cells b. It maintains osmotic balance

c. It speeds up digestion d. It removes excess nitrogen from the body

Activity-Based MCQs :

45. A student places a slice of potato in a concentrated salt solution for a few hours. What happens to the potato cells?

a. They absorb water and become swollen b. They lose water and undergo plasmolysis

c. They divide rapidly d.They remain unchanged

46. A biology teacher asks students to observe an onion peel under the microscope. To clearly see the nucleus, what stain should they use?

a. Iodine solution b. Safranin c. Methylene blue d. Benedict's solution

47. During an experiment, a student places a few grapes in plain water. After a few hours, the grapes swell. What is this an example of?

a. Diffusion b. Osmosis c. Active transport d. Endocytosis

48. A researcher observes cells under a microscope and notices they lack a defined nucleus and membrane-bound organelles. What type of cell is being observed?

a. Eukaryotic cell b. Plant cell c. Prokaryotic cell d. Animal cell

49. A student conducts an experiment where they place a drop of ink in a beaker of water. Over time, the ink spreads evenly throughout the water. What process does this demonstrate?

a. Osmosis b. Active transport c. Diffusion d. Plasmolysis

50. A doctor examines a patient who complains of fatigue and weakness. A blood test reveals low hemoglobin levels. Which function is affected due to this deficiency?

a. Oxygen transport b. Digestion of food c. Water absorption d. Removal of nitrogen waste